

Payout Mechanisms as Operators on Attention Distributions: Identities, Invariances, and Tail Action

Tonify Research

August 2026 · v0.5 (structural pass over v0.4; math untouched) · companion code:
`tonify-sims`

1 Problem statement (verbal)

A streaming platform collects listener money and delivers it to artists. There are essentially three delivery mechanisms. *Pro-rata*: all money goes into one pool and an artist is paid their share of total platform plays — how Spotify and nearly every major service works. *User-centric*: each listener’s wallet is split only across the artists that listener actually played — the perennially proposed “fairer” model. *Direct*: no pool at all; money arrives from devoted fans (donations, artist subscriptions), as on Patreon or Twitch. On top of any mechanism may sit a label contract taking the lion’s share. This paper treats the three mechanisms as operators: input — the table of “who played whom how many times” over a period; output — each artist’s income over the same period.

Four questions are answered. *First*: who gains from a pro-rata $\rightarrow$ user-centric switch, and is it predictable in advance (Theorem 1: audience listening intensity decides, by an exact formula)? *Second*: can a pool-rule reform close the gap between a signed and an independent artist (Lemma 2: the income gap — never; the minimum-viable-audience gap — sometimes)? *Third*: does the direct economy reduce top-end concentration (it does not: it preserves the upper tail and adds a mass of exact zeros at the bottom — inequality is relocated, not reduced)? *Fourth*: is direct income more reliable than the pool (on small audiences it is a lottery, and no parameter values make it unconditionally better — Theorem 4)?

What counts as an answer: a proved statement under explicitly listed assumptions, plus machine verification — every identity and bound is recomputed by a deterministic script whose numbers are quoted in the text. Every claim carries one of three statuses: proved, refuted by counterexample, or conjecture. All four lemmas survived two adversarial red-team passes that did find errors — the correction history is Section 5. *Scope*: accounting statics at a fixed play matrix; equilibrium effects (a rule change altering behavior and hence the matrix) are outside and listed as open. The italicized paragraphs (“Why this matters” / “What it says”) form a complete non-technical route; proofs are set in small type.

2 Conceptual formulation

Entities: artist (payee); listener (paying subscriber); the artist–listener pair (active if played at least once in the period); the play (one stream — the primitive cell); the wallet (a subscriber’s net contribution); the pool (sum of wallets; exists only under pool mechanisms); the contract (a

post-rule multiplier); the superfan (an audience member paying directly; direct economy only); the gift (one direct payment; a superfan makes several per period).

Primary vs derived vs assumed: the play table over a period is the single primary object (a platform log, or a generated matrix); listener intensity, artist plays, audience and streams are measured from it; platform averages and per-artist audience statistics are derived arithmetic; model assumptions (equal wallets, the direct-economy probability model, contract multiplicativity, tail class of audiences) are collected in Section 3.3.

“Artist popularity” is not a latent parameter: audience A_i and streams s_i are measured properties of one specific play matrix over one period. In the synthetic world sim4 they are induced by a generative model calibrated to external anchors — same status: properties of the generated matrix. *The fork carried through the whole paper:* Lemmas L1–L4 (Section 3.4) are statements about ANY fixed play matrix, independent of how it arose; the sim4 generator (Section 4.1) produces one concrete matrix so that identities can be machine-checked at scale. No proof uses the generator. Prior art: the type-model crossover belongs to Alaei, Makhdoumi, Malekian & Pekeč (*Mgmt. Sci.* 2022; their notation is mapped to ours in Table 1); budget balance to Bergantiños & Moreno-Ternero (2025); the tail machinery is cited from Breiman (1965), Jessen & Mikosch (2006), Faÿ et al. (2006), Robert & Segers (2008), Denisov–Foss–Korshunov (2010), Maulik–Resnick–Rootzén (2002); what is new here: the general-matrix identity with the AM–HM frame, the Jensen dispersion premium, the kernel coincidence criterion, the invariance formalization, and the distributional direct-vs-pool comparison.

3 Mathematical formulation

3.1 Notation

Table 1: Notation. Status: M = measured, D = derived, A = assumed, P = model parameter.

Symbol	Type / range	Verbal definition	Period / units	Status
i, N	index; finite set	artist; set (and number) of artists	—	M
u, U	index; finite set	listener; set (and number) of listeners	—	M
p, p_{iu}	$\mathbb{Z}_+^{N \times U}$	play matrix; plays of pair (i, u)	period (year); plays	M
P_u	integer ≥ 1 (A1)	listener u ’s intensity: $\sum_i p_{iu}$	year; plays	D
P_i	integer ≥ 0	artist i ’s plays: $\sum_u p_{iu}$	year; plays	D
T	integer > 0	total platform plays: $\sum_u P_u = \sum_i P_i$	year; plays	D
$\bar{P}$	real > 0	mean intensity T/U	year; plays	D
A_i	integer ≥ 0	audience: number of u with $p_{iu} > 0$	year; listeners	D
s_i	real ≥ 0	artist streams (sim worlds: target yearly volume)	year; plays	M / P
W, W_u	real $> 0; \geq 0$	wallet (net subscriber contribution); unequal-wallet version	year; \$	A (A2)
R	real > 0	the pool: $U \cdot W$	year; \$	D
w_u^i	probability vector	listener u ’s share of artist i ’s plays: p_{iu}/P_i	—	D
m_i, H_i	reals > 0	arithmetic / harmonic mean audience intensity under w^i	year; plays	D
PR_i	real ≥ 0	artist i ’s payout for the period under pro-rata, in wallet units (\$)	year; \$	D

Symbol	Type / range	Verbal definition	Period / units	Status
UC_i	real ≥ 0	artist i 's payout for the period under user-centric, in wallet units (\$)	year; \$	D
D_i, Y_A	random ≥ 0	artist i 's direct-economy payout for the period, \$ (Y_A : at audience A)	year; \$	model
S_i	random, $\text{Bin}(A_i, \sigma)$	number of superfans of artist i	year; people	model
σ	real $\in (0, 1)$	superfan share: probability a listener pays directly	—	A (0.017)
k	integer ≥ 1	gifts per superfan per period	year; payments	A (4)
$G, G_{ft}, \bar{g}$	random > 0 ; real	gift size; gift t of fan f ; mean gift $\mathbb{E}G$	\$	A (lognormal; $\bar{g} = 6.886$)
τ	real $\in (0, 1]$	artist's share of a gift	—	A (0.80)
ρ	real $\in (0, 1]$	contract pass-through to a signed artist	—	D (0.06772)
$F, F_i(p)$	operator	arbitrary pool rule; its payout to artist i	year; \$	model
θ	real > 0	payout threshold (\$13 Stars)	\$	P
A_0	integer > 0	boost threshold in the artist-centric counterexample (15 000)	listeners	P
V	real > 0	target yearly income defining MVA (\$1 200/yr)	year; \$	P
MVA_s, MVA_t	integers > 0	minimum viable audience (signed / independent)	listeners	D
c, rate	reals > 0	pool rate per listener $c = \text{PL} \cdot \text{rate}$; external per-stream anchor	\$/listener/yr; \$/play	D / M
PL	real > 0	plays per active pair (21.21, LFM-1b, caveat CRITIC §2)	year; plays/pair	M
X_A	real > 0	pool income at audience A : cA (deterministic)	year; \$	D
d^*	real > 0	direct-vs-pool breakeven: $\text{PL} \cdot \text{rate}/(\sigma \bar{g} \tau)$ payments/superfan/yr	payments/yr	D
$\varphi(t)$	function, $t > 0$	gift Laplace transform $\mathbb{E}e^{-t\tau G} < 1$	—	D
α, α_G	reals	tail index of audiences; of gifts	—	A (1.4) / M
$L(x)$	function	slowly varying part of $\mathbb{P}(A > x) = x^{-\alpha} L(x)$	—	A
M_i, m^*	random $\in (0, 1]$; real	UC multiplier $\mathbb{E}_{w^i}[1/P_u]$; its tail limit in (C)	1/plays	D / A
q, q_m	reals $\in [0, 1)$	zero mass of direct $\mathbb{E}[(1 - \sigma)^A]$; measured zero share	—	D / M
$G_{\text{pool}}, G_{\text{direct}}, G^+$	reals $\in [0, 1]$	Gini of pool incomes; of direct; among positive incomes	—	D
n, n^+, μ, μ^+	integers; reals	artist counts (all / positive-income); mean incomes (Gini decomposition)	—	D
δ_0, F^+, F_X, F_Y	distributions / CDFs	point mass at zero; CDF of positive part; CDFs of X_A, Y_A	—	notation
$\text{RV}(\alpha), o_p(1)$	classes	regular variation of index α ; vanishing in probability	—	notation
$\ell_i, \tilde{A}_i$	real ≥ 1 ; integer	sim4 target listeners $\max(1, s_i/\text{PL})$; realized audience	year; listeners	D (sim4)

Symbol	Type / range	Verbal definition	Period / units	Status
K_u	integer $\in [1, 2000]$	sim4 listener's yearly playlist size	year; artists	A ($\text{LN}(\ln 12, 1.0)$) P / D
$\gamma, \bar{A}_g$	real; real > 0	sim4 intensity-size coupling dial $\{-0.3, 0, +0.3\}$; geometric-mean audience	—	P / D
$U_{\text{sil}}, U_{\text{act}}, U_{\text{tot}}$	integers	silent, active, all subscribers (without A1)	people	D
ε, a, t, x, z	reals	local variables of limits and integrals	—	notation
ρ_{eff}	real $\in (0, 1]$	effective pass-through under component-heterogeneous contracts: $\sum_m \rho_m w_{im}(F)$	—	D
Δ	operator	change of a quantity (e.g. ΔGini pool $\rightarrow$ direct)	—	notation
Bin, LogNor-mal, corr	standard	binomial and lognormal laws; Pearson correlation	—	notation
<i>Foreign notation (Alaei et al. 2022 type model) and dictionary to ours:</i>				
q_i, λ_i	real; real	mass and intensity of listener type i $\leftrightarrow$ our group with $P_u = \lambda_i$	—	foreign
π_{ij}	real	type i 's preference share for artist j $\leftrightarrow$ the type's play distribution	—	foreign
$\bar{\lambda}$	real	$\sum_i q_i \lambda_i \leftrightarrow$ our $\bar{P}$ on a type population	—	foreign
β_{pr}, β_{uc}	reals	their budget rates; equality holds by construction here (one pool R)	—	foreign

3.2 The three operators: definitions, domains, edge cases

Payout operators map $\mathbb{R}_+^{N \times U} \rightarrow \mathbb{R}_+^N$. *Pro-rata*: $\text{PR}_i = UW P_i / T$ — artist i 's payout for the period in wallet units; defined for $T > 0$; $P_i = 0 \Rightarrow \text{PR}_i = 0$ ($A_i = 0$ implies $P_i = 0$). *User-centric*: $\text{UC}_i = W \sum_u p_{iu} / P_u$ — payout in wallet units; requires $P_u > 0$ for all paying u (Assumption A1); $p_{iu} / P_u \leq 1$ always, so $\text{UC}_i \leq W A_i$: no listener, however light, blows up the payout. *Direct*: $D_i = \tau \sum_{f=1}^{S_i} \sum_{t=1}^k G_{ft}$ with $S_i | A_i \sim \text{Bin}(A_i, \sigma)$, i.i.d. gifts $G \geq 0$, $\mathbb{E}G = \bar{g}$, all moments finite; $\mathbb{E}[D_i | A_i] = \tau \sigma k \bar{g} A_i$; at $A_i = 0$, $D_i \equiv 0$; at $A_i > 0$, $\mathbb{P}(D_i = 0) \geq (1 - \sigma)^{A_i} > 0$ — the atom at zero, the central object of Section 3.4. *Contract*: a multiplicative layer after the rule, $\rho \cdot F_i(p)$; the payout threshold θ is a separate logistics layer (accumulation vs annual burn), treated as breaking channel B3 of Lemma 2.

3.3 Assumptions

A1 (payer activity) Every paying subscriber listens: $P_u \geq 1$. Silent payers are excluded from *both* pools; otherwise budget balance fails by exactly $W \cdot U_{\text{sil}}$ and all ratios acquire the factor $U_{\text{act}} / U_{\text{tot}}$ (red-team counterexample: 2 artists $\times$ 3 active listeners plus 2 silent give $3/5 = 0.6$). *Used by*: L1 (*all parts*), L3 (*UC bound*).

A2 (equal wallets) One subscription price; measured approximation. The unequal-wallet generalization preserves the identity's structure (Section 3.4, Theorem 1 discussion). *Used by*:

L1, L2, L4.

- A3** (integer plays) $p \in \mathbb{Z}_+^{N \times U}$, hence $1/P_u \leq 1$ and $M_i \leq 1$; with real-valued plays the weaker bound $p_{iu}/P_u \leq 1$ survives. *Used by: L3 (one-sided UC tail bound).*
- A4** (fixed matrix; accounting frame) p is fixed: no equilibrium, pool-size, or welfare claims. *Used by: all lemmas.*
- A5** (direct-economy model) $S_i|A_i \sim \text{Bin}(A_i, \sigma)$ with $\sigma \perp A_i$; gifts i.i.d., all moments finite; k deterministic. *Used by: L3 (direct branch), L4.*
- A6** (multiplicative contract, single ρ) The contract multiplies the payout by a constant ρ , uniform across income components and markets; violations are channels B1, B2, B5. *Used by: L2.*
- A7** (rule applied at artist level) F acts at the artist level, the contract strictly downstream; domain $F_i(p) > 0$; violation is channel B6. *Used by: L2 (2a).*
- A8** (heavy-tailed audiences) $\mathbb{P}(A > x) = x^{-\alpha}L(x)$, $\alpha > 1$ (sim1 world: $\alpha = 1.4$, gate T3; empirical caveat Spierdijk & Voorneveld). *Used by: L3.*
- (C)** (tail-conditional concentration; UC branch of L3 only) For every $\varepsilon > 0$, $\mathbb{P}(|M_i - m^*| > \varepsilon | A_i > a) \rightarrow 0$ as $a \rightarrow \infty$, $m^* > 0$. Measurable on full log datasets.

3.4 Lemmas and theorems

The fork of Section 2 stands: everything here is about an arbitrary fixed play matrix; no proof uses the sim4 generator.

L1: the ratio identity. *Uses assumptions A1, A2, A4.*

Why this matters. *Debates about user-centric run on intuition (“it helps niche artists”). The identity below replaces intuition with a formula: an artist’s user-centric-to-pro-rata income ratio is a specific average over that artist’s own audience, so the winners of a reform can be computed before the reform.*

Theorem 1 (Identity). *For any artist i with $P_i > 0$, under equal wallets,*

$$\frac{\text{UC}_i}{\text{PR}_i} = \mathbb{E}_{w^i} \left[\frac{\bar{P}}{P_u} \right], \quad w_u^i = \frac{p_{iu}}{P_i}.$$

Proof. $\text{UC}_i/\text{PR}_i = (W \sum_u p_{iu}/P_u)/(UWP_i/T) = (T/U) \sum_u (p_{iu}/P_i)(1/P_u) = \bar{P} \mathbb{E}_{w^i}[1/P_u]$. □

With $m_i = \mathbb{E}_{w^i}[P_u]$ (arithmetic) and $H_i = 1/\mathbb{E}_{w^i}[1/P_u]$ (harmonic mean of listener intensity under play weights),

$$\frac{\text{UC}_i}{\text{PR}_i} = \frac{\bar{P}}{H_i} = \underbrace{\frac{\bar{P}}{m_i}}_{\text{level}} \cdot \underbrace{\frac{m_i}{H_i}}_{\text{dispersion} \geq 1},$$

the second factor ≥ 1 by AM–HM, with equality iff P_u is constant on the support of w^i . Under unequal wallets W_u the identity becomes $\text{UC}_i/\text{PR}_i = \mathbb{E}_{w^i}[W_u/P_u]/(\sum_u W_u/T)$ — same structure, “intensity” replaced by “plays per dollar”.

What it says. The ratio splits into level $\times$ dispersion: a “light” audience (below-average listeners) helps through the first factor, and a heterogeneous audience always helps through the second. So the folk claim “user-centric helps the niche” is wrong in an instructive way: what helps is not nicheness but lightness and heterogeneity of one’s listeners.

Corollary (a) (Coincidence; the naive converse is false). $P_u \equiv \bar{P}$ implies $UC = PR$ componentwise. The converse fails: with one artist and two listeners, $p = (10, 20)$, intensities differ yet $UC_1 = PR_1 = 2W$. Exact criterion: $UC_i = PR_i$ for all i iff the vector $c_u = 1/P_u - U/T$ lies in $\ker p$.

Corollary (b) (Zero sum). $\sum_i UC_i = \sum_i PR_i = UW$: user-centric is a pure redistribution of the same pool.

Corollary (c) (Crossover). $UC_i > PR_i$ iff $H_i < \bar{P}$. On type populations this reduces to the condition of Alaei, Makhdoumi, Malekian and Pekeč (Mgmt. Sci. 2022, Props. 2, 8): artist j weakly prefers pro-rata iff $\sum_i q_i \pi_{ij} (\lambda_i - \bar{\lambda}) \geq 0$.

Corollary (d) (Dispersion premium). Fix $m_i = \bar{P}$. Then $UC_i/PR_i = m_i/H_i \geq 1$, with equality iff audience intensity is degenerate: at the same mean level, within-audience dispersion strictly raises user-centric income.

L2: pass-through invariance. Uses assumptions A2, A4, A6, A7 (part 2a); A2, A4, A6 and a monotone income with threshold V (part 2b).

Why this matters. The central policy dispute: is the artist squeezed by the pool rule or by the label contract? If a signed artist keeps seven cents on the dollar, can any reform of the splitting rule close that gap? The lemma answers: never in income terms — and sometimes in minimum-viable-audience terms.

Lemma 2 (Two-part invariance). (2a) For any pool operator F applied at the artist level, with the multiplicative contract strictly downstream and on the domain $F_i(p) > 0$: $\rho F_i/F_i = \rho$ — the signed/independent income gap is invariant to the rule (a nonlinear F applied to a licensor’s catalogue breaks this). (2b) The minimum-viable-audience gap satisfies $MVA_s/MVA_t = 1/\rho$ iff $F^{-1}(V/\rho) = F^{-1}(V)/\rho$; linearity of F -income in audience is sufficient but not necessary (a strictly increasing log-periodic F is nonlinear yet yields $1/\rho$ for all V). A Deezer-style $\times 2$ boost above a threshold yields $1/(2\rho)$ instead (counterexample in companion code) while the income ratio stays ρ pointwise.

What it says. A rule reform cannot move the contractual income gap by one cent; it can move the “minimum audience to make a living” gap, but only by breaking the scaling criterion above. This formalizes the empirical regularity that label shares did not move across rule changes (Pedersen 2020). Breaking channels: endogenous $\rho(F)$; additive components; payout thresholds (accumulation delays $\times 1/\rho$ in time); behavioral endogeneity of p ; component/market-heterogeneous $\rho_{\text{eff}} = \sum_m \rho_m w_{im}(F)$; payee-level aggregation of a nonlinear rule.

L3: tail action and the atom at zero. Uses assumptions A1, A3, A4, A8; condition (C) — UC branch only; A5 — direct branch only.

Why this matters. Streaming’s chief indictment is concentration: the top percent takes nearly everything. Would a direct fan-to-artist economy spread income more evenly? This section answers for the top and the bottom of the distribution separately — and the answer disappoints every mechanism.

Let $\mathbb{P}(A > x) = x^{-\alpha}L(x)$. PR is linear: index preserved. UC: $UC_i = WP_i \mathbb{E}_{w^i}[1/P_u]$ with bounded multiplier; under tail-conditional concentration of the multiplier the index is preserved, and heavier tails are impossible for any dependence. Direct: binomial thinning plus a light-tailed compound sum preserve α (Jessen–Mikosch 2006, Lem. 3.7; Faÿ et al. 2006; Denisov–Foss–Korshunov 2010; Breiman 1965 enters only through the multiplicative contract layer), while $\mathbb{P}(D_i = 0 | A_i) \geq (1 - \sigma)^{A_i} \geq e^{-1}(1 - \sigma)$ for $A_i \leq 1/\sigma$: an *atom at zero* on small audiences. With zero mass q , $G_{\text{direct}} = q + (1 - q)G^+$: the mechanism relocates inequality from the intensive to the extensive margin; no mechanism softens the upper tail.

What it says. No mechanism touches how much the top takes; what changes is the shape of the bottom. Pool rules pay many tiny positive amounts; the direct economy converts most of those into exact zeros — at the calibrated superfan share, at least a third of small-audience artists earn literally nothing. Inequality is relocated, not reduced.

L4: dominance blockade. Uses assumptions A2, A4, A5; the anchor $c = \text{PL-rate}$ (independent-artist comparison).

Why this matters. Suppose the direct economy pays four times the pool on average. Should a small artist leave the pool? Not necessarily: the average hides a large chance of zero. This section compares the two incomes by their whole distributions, in the standard language of stochastic dominance.

Fix A with $(1 - \sigma)^A > 0$ and pooled income $X_A = cA > 0$ (deterministic given A); $Y_A = D_A$ as above.

Theorem 4 (Full blockade). (i) Y_A never first-order dominates X_A ; (ii) Y_A never second-order dominates X_A , for any parameters; (iii) $X_A \succeq_2 Y_A$ iff $\mathbb{E}Y_A \leq X_A$, i.e. iff $k \leq d^* = \text{PL-rate}/(\sigma\bar{g}\tau)$; (iv) for $k > d^*$ the pair is incomparable in both orders.

Proof. (i) For $x \in (0, X_A)$: $\mathbb{P}(X_A > x) = 1$ while $\mathbb{P}(Y_A > x) \leq 1 - (1 - \sigma)^A < 1$. (ii) SSD against degenerate X_A requires $\int_0^t F_Y \leq \max(0, t - X_A)$; at $t = X_A$ the left side is $\geq (1 - \sigma)^A X_A > 0$. (iii) $(\Leftarrow)$ $\int_0^t F_Y = t - \mathbb{E} \min(Y, t) \geq t - \mathbb{E}Y \geq t - X_A$ and ≥ 0 . $(\Rightarrow)$ $t \rightarrow \infty$ gives $\mathbb{E}Y \leq X_A$. (iv) From (i)–(iii). $\square$

Remark. The correct comparison is mean versus risk: for $k > d^*$ direct pays an expectation premium priced by small-audience lottery risk; $\mathbb{P}(Y_A < X_A) \leq \inf_{t>0} e^{tX_A} (1 - \sigma + \sigma\varphi(t)^k)^A$ with $\varphi(t) = \mathbb{E}e^{-t\tau G} < 1$ — exponential decay in A . In the companion sim1 calibration at $A=30$, $k=4$: mean advantage $\times 4$, yet $\mathbb{P}(Y < X) \approx 0.60$ — and on the full 2M-sample run $\mathbb{P}(Y < X) - \mathbb{P}(Y = 0) = 0$ to machine precision: losing to the pool is exactly the zero outcome. Under the contract layer $c = \text{PL} \cdot \text{rate} \cdot \rho$ the threshold scales to $\rho d^* \approx 0.068$: for a signed artist, direct wins in expectation already at $k \geq 1$.

What it says. The zero atom blocks stochastic dominance in both directions and in both orders, for all parameter values: the direct economy can never be unconditionally better than the pool, no

matter how generous the fans. The honest frame is mean versus risk — direct pays an expectation premium priced by a lottery on small audiences, and only raising the superfan conversion σ (not payment frequency k) attacks the lottery itself.

4 Numerical implementation

Generative model (sim4). Artist popularity: piecewise $s_i \sim$ (lognormal body / log-bridge / Pareto($\alpha=1.4$) tail), calibrated to anchors T_1 : 87% of artists below 10^3 streams/yr, T_2 : 2.6% above 225,734, T_3 : top 0.28% holding 40–55% of streams (gate band; the seed-42 matrix realizes 40.3%, sim1’s larger world 44.5%); target audiences $\ell_i = s_i/\text{PL}$, $\text{PL} = 21.21$ plays per pair (LFM-1b). Playlist sizes $K_u \sim \text{LogNormal}(\ln 12, 1.0)$ (assumed; an MVP-measured dial). Artist selection: exact weighted sampling without replacement, $\mathbb{P}(i \in S_u) \propto \ell_i$ (Gumbel top- K_u), i.e. preferential attachment. Pair intensity, three named regimes on one graph:

(a) ergodic: integer p_{iu} , $P_u \equiv 2048$; (b) $p_{iu} = \lceil \text{LN}(\ln 5.16, 1.676) \rceil$ rescaled to s_i ; (c) $p_{iu} \propto \text{LN} \cdot (A_i/\bar{A}_g)^\gamma$.

Regime (a) has $P_u \equiv 2048$ exactly (a power of two: every p_{iu}/P_u is a dyadic rational, so the control identity is exact in floating point) and is the control (Corollary (a)); γ dials the intensity–size coupling.

Machine verification (deterministic, seed 42; PASS/FAIL gates printed before any conclusion, FAIL exits with code 1): the Theorem-1 identity and AM–HM decomposition on 200 random matrices (max error 8.9×10^{-16}), on the full 21.8M-pair sim4 matrix (7.1×10^{-15}), and — since August 2026 — on *real* Last.fm logs (31,040 artists, 1.02M pairs: 5.8×10^{-15} , with $\text{AM} \geq \text{HM}$ holding artist by artist); zero-sum (1.8×10^{-14}); world anchors T1–T3; the exact-zero control gate ($\max |\text{UC}/\text{PR} - 1| = 0$); the L4 atom at $A=30$ (0.598 vs theory 0.598, 2M samples) with $\mathbb{P}(Y < X) - \mathbb{P}(Y = 0) = 0$; the calibrated-world Gini identity $q_m + (1 - q_m)G^+ = G_{\text{direct}} = 0.973844$ to machine zero ($G_{\text{pool}} = 0.972$; robust to lognormal gifts); both L2 counterexamples (MVA 12 772 / 94 296, ratio $1/(2\rho)$); the log-periodic F giving exactly $1/\rho$; crossover-sign gate: zero violations of Corollary (c) across all artists and regimes.

Reproducibility: python3/numpy/scipy/matplotlib, no external services. `git clone https://github.com/Pr`
`python3 paper/theory_check.py` (seconds; CHECK 1–6); `python3 sim4/bipartite_gen.py`
 (~7 min; gates G1–G3, exports, figures 15–16); `python3 run_all.py` (everything). All numbers quoted above are outputs of these scripts.

5 Red team: two adversarial passes

Two independent red-team passes attacked every lemma (recompute every inequality, reproduce every derivation, hunt counterexamples). Pass one: five attackers, verdict “sound with fixes” in all five zones — nine confirmed lemma findings and ten consistency findings. Pass two: five fresh attackers filed 25 accusations, all 25 confirmed by independent verification. No lemma broke in either pass. Retracted: one inequality (the atom bound misprint, corrected to $e^{-1}(1 - \sigma)$), one concentration bound (Hoeffding, violated $\sim 45\times$ by random gifts; replaced by the Chernoff bound via $\varphi(t)$), the necessity claim “only if linear” in (2b) (log-periodic counterexample; exact criterion $F^{-1}(V/\rho) = F^{-1}(V)/\rho$ installed), and one simulation narrative (head saturation 5.00 — circular sampling; the real head has $\tilde{A}/\ell = 1.08$). Introduced: load-bearing assumption A1 (silent subscribers), breaking channels B5 and B6, the artist-level assumption A7. Full protocols live in

the working-session journals outside the repo; the Russian THEORY.md carries the episode-by-episode narrative.

6 Open questions

(1) Equilibrium versions of L1/L2 (endogenous p); does the Jensen premium survive? (2) The sign of ΔGini pool $\rightarrow$ direct in general (conjecture: a threshold $\sigma^*(\alpha, k)$ exists for the positive-income Gini, never for the full one). (3) Exact characterization of when UC thins the tail (Maulik–Resnick–Rootzén). (4) L4 for recurrent k with churn (dominance on income flows). (5) An axiomatic characterization of direct alongside the B&MT axiomatizations of PR and UC.

7 Empirical bridge

Each lemma rests on a measurable quantity: k (payments per superfan per year; benchmarks Twitch 12, Patreon 12, breakeven median 1.25); $\sigma(A)$ (superfan conversion vs audience size — tests A5); intensity–size coupling across artist sizes (closes condition (C)) — **now measured**: on Last.fm logs the reduced pair-intensity slope is $+0.101 \pm 0.004$ overall, $+0.232$ at the head, $+0.023$ among beyond-mainstream listeners, decomposing into retention $+0.254$ and an *anti*-coupled per-day rate -0.154 (companion **emp1**); the gift tail α_G (whales vs attention regime); $\rho(F)$ (pass-through endogeneity, channel B1 — the Deezer artist-centric switch is a natural experiment); the threshold θ and its censoring mode (channel B3).